Ayman Rimah Said

Laboratoire Mathématiques de Reims, 51100 Reims.
ayman.said@cnrs.fr
aymanrimah.github.io

Current employment

10/24– **Chargé de recherche au CNRS**, *Laboratoire Mathématiques de Reims*, Reims, France

Academic professional experience

8/23–9/24 **Research Associate**, *Cambridge University*, Cambridge, United Kingdom
7/21 – 7/23 **Griffiths Research Assistant Professor**, *Duke University*, Durham, NC, USA
9/18–6/21 **PhD**, *ENS Paris Saclay*, *Paris*, France
9/19–9/20 **Teaching Assistant**, *Université de Paris Sud*, *Paris*, France
9/18–9/19 **Teaching Assistant**, *ENS Paris Saclay*, *Paris*, France

Studies

9/18–6/21 **ENS Paris Saclay-Laboratoire de Mathématiques d'Orsay**, *Paris, France*
Thesis
9/17–6/18 **École Polytechnique-Université de Paris Sud- Paris Saclay**, *Paris, France*
M2 Analyse, Arithmétique, Géométrie.
9/14–3/17 **École Polytechnique**, *Paris*
9/13–7/14 **Classes préparatoires of lycée Hoche**, *Versailles, France*
Mathematics and Physics (MP*).

Current Funding

9/25–9/30 **Member of the ANR project SMASH**

Prior Funding

8/23–9/24 **Research Associate at Cambridge University**, *funded by UKRI grant SWAT.*
8/21–8/23 **Phillip Griffiths Assistant Research Professor of Mathematics at Duke University**, *Partially funded by NSF grants DMS-2043024 and DMS-2124748.*
9/18–9/21 **Doctoral Grant AMX, École Polytechnique**

Published Articles

1. **A Classification Theorem for Steady Euler Flows**, *Tarek M. Elgindi, Yupei Huang, and Chunjing Xie*, To appear in Duke Mathematical Journal,2026
2. **Wellposedness and singularity formation beyond the Yudovich class**, *with T. M. Elgingi and R. M. Murray*, Journal of the European Mathematical Society,2026
3. **A unified approach to mixing and regularity for passive scalar transport by Sobolev vector fields**, *with L. Huysmans*, Journal of Functional Analysis, 2026
4. **On the long-time behavior of scale-invariant solutions to the 2d Euler equation and applications**, *with T. M. Elgindi and R. M. Murray*, Les Annales de l'ENS, 2025
5. **Logarithmic spirals for 2d perfect fluids**, *with I-J. Jeong*, Journal de l'École polytechnique — Mathématiques, 2024
6. **A geometric proof to the Quasi-linearity of the Water-Waves system**, SIAM Journal on Mathematical Analysis, 2023

7. **On the Cauchy problem for dispersive Burgers type equations**, Indiana University Mathematics Journal, 2023
8. **On Paracomposition and change of variables in Paradifferential operators**, Journal of Pseudo-Differential Operators and Applications, 2023
9. **Regularity results on the flow map of periodic dispersive Burgers type equations and the Gravity-Capillary equations**, Water Waves, 2023

Preprints in the peer review process

10. **Small scale creation of the Lagrangian flow in 2d perfect fluids**, arXiv preprint, arXiv:2401.06476
11. **Mixing Estimates for Passive Scalar Transport by BV Vector Fields**, *with L. Huysmans*, arXiv preprint, arXiv:2504.03023
12. **Generic small-scale creation in the two-dimensional Euler equation**, *with T. Alazard*, arXiv preprint, arXiv:2603.13079
13. **On smooth inviscid vortices with fat tails**, *with P. Raphaël*, arXiv preprint, arXiv:2609.04786